Extensions of the Hilbert-space formalism for full FFGFT

Four mathematical structures, one geometric source

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Preamble

Document 230 showed how FFGFT statements can be *translated into the standard Hilbert space* – with reductions losing information at each stage. This document reverses the question:

> *"What additional mathematical structures must be added to the standard Hilbert space so that it can carry the full FFGFT geometry* ***natively*** *– without reduction?"*

The answer has a notable property: each required extension corresponds to an *established mathematical structure* from other areas of theoretical physics. FFGFT does not "invent" new mathematics; it is the specific combination of established extensions that delivers all Standard-Model constants parameter-free.

Notation

Geometric spaces and Hilbert spaces

Symbol	**Meaning**
$\mathbb{R}$	real numbers
$\mathbb{C}$	complex numbers
$\mathbb{C}^2$	standard qubit Hilbert space
$\mathbb{Z}$	integers (winding index)
$\aleph_0$	countably infinite (Hilbert-space dimension after extension 2)
S^1, S^2	circle, 2-sphere
T^n	n-torus, n compact loops
$L^2(M, S)$	sections of spinor bundle S over manifold M
$\mathcal{H}_{\text{std}}$	standard Hilbert space
$\mathcal{H}_{\text{FFGFT}}^{T^n}$	FFGFT Hilbert space on T^n
$\bigoplus_n$	direct sum over index n
$\bigotimes$	tensor product

Algebraic structures

Symbol	Meaning
$SU(2)$	special unitary group (standard)
$SU_q(2)$	q-deformed quantum group (Drinfeld-Jimbo)
q	deformation parameter, $q = e^{i\pi(1-K_{\mathsf{frak}})} = e^{i\pi/75}$
$[n]_q$	q-deformed number, $[n]_q = (q^n - q^{-n})/(q - q^{-1})$
$\sigma_x, \sigma_y, \sigma_z$	Pauli matrices
$\sigma_x^{\mathsf{FFGFT}}$	FFGFT-modified $\sigma_x = K_{\mathsf{frak}}\sigma_x$
$[A, B]$	commutator $AB - BA$
$\{A, B\}$	anticommutator $AB + BA$
$\mathbb{1}$	identity matrix
$\delta_{ij}, \epsilon_{ijk}$	Kronecker delta, Levi-Civita tensor

FFGFT geometry parameters

Symbol	Meaning
$\xi_0 = 4/30000$	dimensionless FFGFT geometry parameter
ξ	alternative notation for ξ_0
$D_f = 2.999867$	fractal spacetime dimension
K_{frak}	fractal correction, $K_{\mathsf{frak}} = 1 - 100\xi_0 \approx 0.9867$
$1 - K_{\mathsf{frak}}$	deformation phase, ≈ 0.0133
v	Higgs vacuum expectation value, ≈ 246 GeV
$\hbar, c$	Planck constant, speed of light

Winding structure (extensions 2 and 4)

Symbol	**Meaning**
$n \in \mathbb{Z}$	cyclic winding quantum number (loop)
$\|\psi; n\rangle$	state with winding number n
$\mathbb{C}^2_n$	qubit Hilbert space in n-winding sector
E_n	energy of n-th winding excitation
R	loop radius (major radius of torus)
$a^\dagger, a$	creation and annihilation operators for winding excitations
$k_t \in \mathbb{Z}$	temporal winding quantum number
$f(k_t)$	periodic energy function of k_t winding
(n_x, n_y, n_z)	spatial winding quantum numbers
(n_θ, n_ϕ)	spin winding quantum numbers

Fiber bundle operators (extension 3)

Symbol	**Meaning**
$\vec{L}$	orbital angular momentum operator
$\vec{S}$	spin operator
L_i, S_j	components of $\vec{L}, \vec{S}$
H_{SO}	spin-orbit coupling Hamiltonian
$\xi(r)$	radial spin-orbit coupling function
Z	atomic number
S	spinor bundle over T^3 or T^4
$\mathcal{D}_{T^4}$	Dirac-like operator on T^4

Standard QM quantities

Symbol	Meaning
$\|\psi\rangle$	quantum state
$\|0\rangle, \|1\rangle$	computational basis states
α, β	complex amplitudes
$\psi(x, y, z)$	position wavefunction
$\omega = 2\pi/T$	angular frequency
E	energy
m, m_i	mass, mass of fermion i
α	fine-structure constant
∇^2	Laplace operator

The four required extensions

From Doc. 230 we know: the Hilbert-space translation of a single qubit takes place on the Bloch sphere (S^2), reached through two reductions from the full FFGFT geometry ($T^4 \to T^3 \to T^2 \to$ cylinder $\to S^2$). Each reduction corresponds to the loss of a structural layer. To reconstruct full FFGFT, we must restore four extensions.

Overview:

Step	Extension	Mathematical model
$S^2 \to$ cylinder	deformed $SU(2)$ algebra	quantum groups $SU_q(2)$
cyl. $\to T^2$	additional cyclic quantum number	Kac-Moody algebras, winding states
$T^2 \to T^3$	bundle structure instead of tensor product	fiber bundles, Kaluza-Klein
$T^3 \to T^4$	temporal winding dimension	Kaluza-Klein, compact extra dimensions

1 Extension 1: deformed $SU(2)$ algebra

1.1 Standard Pauli algebra

The Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ generate the isotropic $SU(2)$ algebra:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k, \qquad \{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbb{1}. \tag{1}$$

"Isotropic" means: all three axes x, y, z are on equal footing; a rotation around σ_x is exactly equivalent to a rotation around σ_z after an appropriate basis change.

1.2 FFGFT anisotropy

In FFGFT there is a *distinguished prefactor* for the σ_x rotation:

$$\sigma_x^{\mathsf{FFGFT}} = K_{\mathsf{frak}} \cdot \sigma_x, \qquad K_{\mathsf{frak}} = 1 - 100\xi_0 \approx 0.9867. \tag{2}$$

The σ_y and σ_z rotations are unmodified. Hence the algebra is no longer isotropic: σ_x is distinguished.

1.3 Mathematical identification: $SU_q(2)$

The $SU_q(2)$ quantum group (Drinfeld-Jimbo, 1985) is the q-deformation of $SU(2)$, with modified commutation relations. In the limit $q \to 1$ it becomes standard $SU(2)$. The FFGFT anisotropy corresponds to the identification:

$$\boxed{q = e^{i\pi(1-K_{\mathsf{frak}})} = e^{i\pi\cdot 100\xi_0} = e^{i\pi/75} \approx 0.999 + 0.042i.} \tag{3}$$

The q-phase is small ($\pi/75 \approx 0.042$ rad) but structurally significant: it breaks the spherically symmetric $SU(2)$ algebra into an anisotropic, axis-distinguished algebra. The anisotropy is the mathematical memory of the cylinder of its toroidal origin.

1.4 What this extension costs

- Hilbert space remains $\mathbb{C}^2$ – no dimensional extension.
- Operators remain 2×2 matrices – no matrix extension.
- What changes: the structure constants of the algebra (Casimir operators, multipliers).
- Consequence: all π-gates deviate systematically by $0.013\,\%$ from the ideal value (Doc. 175 §3, testable in high-precision quantum gate experiments).

2 Extension 2: cyclic quantum number

2.1 What is missing on the cylinder

The cylinder $\mathbb{R} \times S^1$ has *one* compact loop (the phase θ) and *one* linear axis (z). On the 2-torus $T^2 = S^1 \times S^1$, *both* axes are compact – so $z = +1$ and $z = -1$ are connected by a second loop. In the Bloch-sphere limit (all poles contracted) and even in the cylinder limit (one loop pulled apart), this connection is lost.

2.2 Hilbert-space extension

To represent the second loop, the Hilbert space must carry an additional *winding quantum number* $n \in \mathbb{Z}$. Instead of

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \in \mathbb{C}^2 \tag{4}$$

the state becomes

$$|\psi; n\rangle = \alpha|0; n\rangle + \beta|1; n\rangle, \tag{5}$$

where n specifies how often the mode configuration wraps around the second loop of T^2. The Hilbert space becomes:

$$\boxed{\mathcal{H}^{T^2}_{\mathsf{FFGFT}} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}^2_n.} \tag{6}$$

This is a *countably infinite-dimensional* Hilbert space, no longer 2-dimensional.

2.3 Low-energy limit: why this is normally invisible

Higher winding excitations ($n \neq 0$) have higher energy:

$$E_n \sim n^2 \cdot \frac{\hbar^2}{mR^2}, \tag{7}$$

where R is the loop radius. At low energies only $n = 0$ is populated, and the Hilbert space effectively reduces to $\mathbb{C}^2$ – the standard Bloch sphere. Higher n-excitations become rapidly invisible at macroscopic R.

2.4 Mathematical model

The structure $\bigoplus_n \mathbb{C}^2_n$ is exactly what appears in string theory as *winding states of compact dimensions*, and in Kac-Moody algebras as *levels*. Here too the mathematics is known – FFGFT imports it with a specific geometric meaning.

2.5 What this extension costs

- Hilbert-space dimension becomes $\aleph_0$ instead of 2.
- Operators become infinite-dimensional, but block-diagonal: within each n-sector they are 2×2.
- Additional operators $a^\dagger, a$ propagate between winding sectors ($n \to n \pm 1$).
- Irrelevant at low energies; required in the full theory (e.g. for vacuum Casimir effects between the poles, Doc. 091/220).

3 Extension 3: bundle structure instead of tensor product

3.1 Standard QM for a particle with spin

A spin-$\frac{1}{2}$ particle in space has, in standard QM, the Hilbert space:

$$\mathcal{H}_{\text{std}} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2. \tag{8}$$

The *tensor product* means: position space and spin space are **independent** Hilbert spaces. Operators on one factor commute with operators on the other:

$$[L_i, S_j] = 0 \qquad \text{(orbital and spin angular momentum commute).} \tag{9}$$

Spin-orbit coupling must be introduced as an *additional Hamiltonian term*:

$$H_{\text{SO}} = \xi(r)\, \vec{L} \cdot \vec{S}, \tag{10}$$

where the "spin-orbit constant" $\xi(r)$ is a free parameter (typically $\sim Z^4\alpha^2$ for a hydrogen-like atom).

3.2 FFGFT: spin and position on the same T^3

In FFGFT both "position" and "spin" arise from windings on the same 3-torus T^3 (Doc. 210):

- spatial windings (n_x, n_y, n_z): mode-space position,

- spin windings (n_θ, n_ϕ): 3-1 spin index.

Both sets of winding numbers live on the same compact manifold. They are not independent.

3.3 Hilbert space as spinor bundle

Mathematically the FFGFT Hilbert space for a particle with spin on T^3 is *not* a tensor product, but the space of sections of a spinor bundle:

$$\boxed{\mathcal{H}^{T^3}_{\mathsf{FFGFT}} = L^2(T^3, S),} \tag{11}$$

where S is the spinor bundle over T^3. In this structure orbital and spin operators are *no longer independent*:

$$[L_i, S_j] \neq 0 \qquad \text{with corrections } \sim \xi. \tag{12}$$

3.4 Consequence: geometric spin-orbit coupling

Spin-orbit coupling in FFGFT is *not* an additional Hamiltonian term, but a **geometric property of the bundle**. The "spin-orbit constant" $\xi(r)$ is not a free parameter, but follows from the winding geometry. In particular: the behaviour $\xi(r) \propto Z^4\alpha^2$ in heavy atoms follows from the specific winding topology (Doc. 174).

3.5 Mathematical model

The bundle structure is standard differential geometry – every modern differential geometry textbook treats spinor bundles. The specific FFGFT step is the geometric identification: "spin and position are not two Hilbert spaces, but two aspects of the same winding structure on a compact manifold."

3.6 What this extension costs

- Hilbert space is *no longer* a tensor product – the algebraic structure differs.

- Pauli principle becomes geometric: two fermions cannot occupy the same winding state because the bundle geometry forbids it (Doc. 174 §3).
- Angular momentum algebra is intertwined with translation algebra.
- Consequence: all phenomenological "spin-orbit constants" of atomic physics are derivable from FFGFT – they are not free.

4 Extension 4: temporal winding k_t

4.1 What standard QM does with time

In standard QM, time is a *continuous, linear* axis $\mathbb{R}$. The Schrödinger equation:

$$i\hbar\frac{\partial\psi}{\partial t} = H\psi, \qquad E = \hbar\omega \qquad \text{(standard dispersion).} \tag{13}$$

Energy is linear in the frequency $\omega = 2\pi/T$.

4.2 FFGFT: time on S^1

In FFGFT, time is a *compact* winding direction with quantum number $k_t \in \mathbb{Z}$ (Doc. 210). The energy becomes:

$$\boxed{E = \hbar f(k_t),} \tag{14}$$

where f is a periodic function. In the lowest winding sector this reduces to $E \approx \hbar\omega$, but the winding structure is real and carries physical meaning.

4.3 Mass from winding

The central FFGFT statement: **the mass of a mode is the k_t-winding energy**:

$$m_i c^2 = \hbar f(k_t^{(i)}) \qquad \text{for mode } i. \tag{15}$$

Combined with the Yukawa form (Doc. 006):

$$m_i = r_i \cdot \xi_0^{p_i} \cdot v, \tag{16}$$

all 12 Standard-Model fermion masses follow from the tuples $(n_x, n_y, n_z, k_t^{(i)})$ without free parameters.

4.4 Hilbert-space extension

Instead of $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ and time as a continuous parameter, the full FFGFT Hilbert space becomes:

$$\mathcal{H}^{T^4}_{\text{FFGFT}} = L^2(T^4, S), \tag{17}$$

with sections of a spinor bundle over the 4-torus. The Schrödinger equation becomes a *mode equation* on T^4:

$$i\hbar \frac{\partial \varphi_{n,l,j}}{\partial t} = \mathcal{D}_{T^4} \varphi_{n,l,j}, \tag{18}$$

where $\mathcal{D}_{T^4}$ is a Dirac-like operator on the 4-torus. At low energies this reduces to the standard Schrödinger equation with fixed mass.

4.5 Mathematical model

Compact extra dimensions have been an established framework since Kaluza (1921) and Klein (1926). FFGFT uses this structure with two modifications:

- The extra dimension is *temporal*, not spatial.
- The winding quantum number k_t is not free, but fixed by geometry and Higgs matching.

4.6 What this extension costs

- The Schrödinger equation itself is modified.
- Masses are no longer external parameters.
- The standard QFT construction (Lagrangian densities with Yukawa terms) is a *low-energy approximation* of the full FFGFT mode theory.

- Consequence: all 12 fermion masses, the fine-structure constant, and the Higgs self-coupling follow parameter-free from the 4D winding geometry.

5 The structural point: known mathematics, new combination

Each of the four extensions corresponds to a mathematically *well-established* structure:

Ext.	**Mathematics**	**Established since**
1	quantum groups $SU_q(2)$	Drinfeld/Jimbo 1985
2	winding excitations	string theory, Kac-Moody (1980s)
3	spinor bundles	differential geometry 1950s
4	compact extra dimension	Kaluza 1921, Klein 1926

None of these structures is new. What makes FFGFT *specific* is the **combination**: all four extensions together, with a single geometric justification.

What no other theory does:

- Quantum groups are studied in mathematics, but not motivated by geometry from a spacetime dimension.
- String theory has winding excitations, but requires 10 or 11 dimensions and does not deliver Standard-Model masses parameter-free.
- Spinor bundles are standard geometry, but typical QFT applications take spin and position as a tensor product.
- Kaluza-Klein theories exist in numerous variants, but none delivers all 12 SM masses from a single compact dimension.

FFGFT is the *convergence* of these four extensions with a common geometric source (T^4 and ξ_0).

6 What a Hilbert-space-extended formulation achieves

If one equips the standard Hilbert space with all four extensions, one obtains a complete FFGFT-equivalent formulation accessible to mathematicians and QM practitioners:

$$\mathcal{H}^{\mathsf{FFGFT}} = L^2(T^4, S) \bigotimes (\text{deformed } SU_q(2) \text{ algebra}), \quad (19)$$

with an additional winding quantum number n per spin rotational degree of freedom and a modified mode equation $\mathcal{D}_{T^4}\varphi = E\varphi$ instead of the Schrödinger equation.

In this formulation all FFGFT predictions are derivable directly from the Hilbert-space algebra:

- masses from the k_t windings,
- fine-structure constant from the $SU_q(2)$ Casimir,
- spin-orbit coupling from the bundle structure,
- K_{frak} corrections from the quantum-group deformation,
- Casimir effect and cosmological energy loss from the winding vacuum structures.

Methodological consequence: FFGFT can be *fully formulated in Hilbert-space language*, provided one accepts the four extensions. Those wishing to extend the standard Hilbert space minimally can begin with stage 1 ($SU_q(2)$), then stage 2 (winding quantum number), and so on – each stage brings a further class of FFGFT statements into the Hilbert-space vocabulary.

7 Summary

The standard Hilbert space ($\mathbb{C}^2$ per qubit, $L^2(\mathbb{R}^3)$ per particle, tensor product for composite systems) must be supplemented by **four extensions** to capture full FFGFT:

1. **Deformed algebra** $SU_q(2)$ with $q = e^{i\pi(1-K_{\mathsf{frak}})}$.
2. **Cyclic winding quantum number** $n \in \mathbb{Z}$ for each compact dimension; Hilbert space becomes $\bigoplus_n \mathbb{C}_n^2$.
3. **Bundle structure** instead of tensor product between position and spin: $L^2(T^3, S)$ with coupled operators.
4. **Temporal winding dimension** k_t with energy $E = \hbar f(k_t)$ instead of $E = \hbar\omega$; masses from k_t windings.

Each extension corresponds to an established mathematical structure (quantum groups, string-theory windings, differential geometry, Kaluza-Klein). FFGFT is the specific, geometrically motivated combination of these four structures, delivering all Standard-Model constants parameter-free.

Methodological position:

- FFGFT *integrates* into the Hilbert-space formalism, provided it is appropriately extended.
- The extensions are *not ad hoc*, but follow from the specific geometric structure of T^4.
- Mathematicians and QM practitioners can adopt FFGFT *stage by stage* – each extension turns one further class of previously external parameters into derivable quantities.

Connection to Doc. 230: while Doc. 230 described the “downward translation” (FFGFT into the standard Hilbert space, with reductions), this document formulates the “upward translation” (Hilbert-space extensions, to carry FFGFT natively). Both directions are necessary to fully understand the structural connection.